1 A student is investigating the motion of magnets falling through a vertical copper pipe as shown in Fig. 1.1.

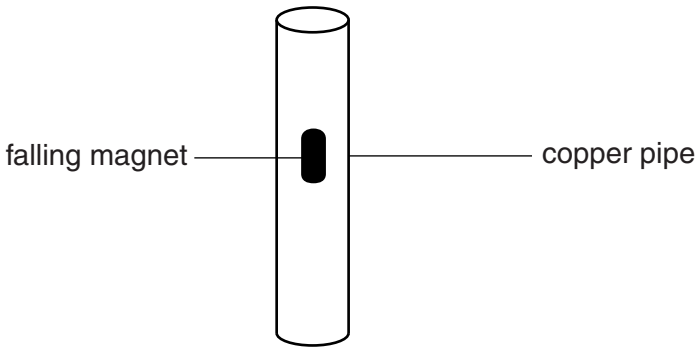

Fig. 1.1

The student releases a magnet above the copper pipe. The magnet has speed v as it leaves the pipe.

It is suggested that the relationship between v and B is

$$v = v_0 e^{-\lambda B}$$

where B is the magnetic flux density at the poles of the magnet and v_0 and λ are constants.

Design a laboratory experiment to test the relationship between v and B. Explain how your results could be used to determine values of v_0 and λ. You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

- the procedure to be followed,
- the measurements to be taken,
- the control of variables,
- the analysis of the data,
- any safety precautions to be taken.

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